Spaceflight transcriptome responses in *Arabidopsis thaliana* are largely study-specific: a study-aware re-analysis of NASA OSDR data with fluid dynamics of the flight growth hardware

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Abstract

Spaceflight transcriptome studies of *Arabidopsis* differ in hardware, tissue, platform and flight, and their results rarely agree. We re-analysed 307 samples from 11 NASA Open Science Data Repository (OSDR) studies with study-aware models, and coupled them to lattice-Boltzmann computational fluid dynamics (CFD) of the BRIC, CARA and VEGGIE growth hardware. A flight response shared by all studies explained under 1% of gene variance (0.3% pooled, under 0.01% in RNA-seq alone), against 7–11% for study-specific flight effects. Within BRIC, flight signatures agreed between the RNA-seq studies (median $r = 0.40$) but not with the earlier microarray studies, which detected almost no flight response. CFD-predicted boundary-layer conductance and carbon supply were confounded with study and did not explain which studies responded: in study-aware models across all studies, no pathway showed a flight effect or a CFD dose-response. Consistent with this, the CFD shows that inside the flight dishes gravity changes leaf gas exchange only about 1.4–1.8-fold, against 2.25-fold in open air, and that sealed BRIC dishes become carbon-limited on the ground as well as in orbit. Spaceflight responses are real but largely experiment-specific, which argues for replicated flights in common hardware rather than pooled meta-analysis.

Keywords: spaceflight, *Arabidopsis*, transcriptomics, meta-analysis, study heterogeneity, computational fluid dynamics, boundary layer, NASA OSDR

Introduction

Spaceflight alters gene expression in *Arabidopsis* [Paul et al., 2017], and the NASA OSDR now holds transcriptomes from many flight experiments. These experiments used different growth hardware: sealed Petri-dish canisters (BRIC), semi-permeable dishes (CARA) and vented chambers with forced airflow (VEGGIE). They also differ in tissue, ecotype, lighting, plant age and measurement platform. Pooling them promises statistical power, but only if a common flight response exists to be found.

Growth hardware also sets the atmosphere around the plant. In microgravity buoyant convection is lost, so air movement around a leaf depends on ventilation, and forced airflow can restore gas exchange [Kitaya et al., 2003, Monje et al., 2005]. If hardware-dependent gas exchange shaped the flight response, quantities predicted by CFD of each hardware, such as the leaf boundary-layer conductance g_{bl}, should explain which experiments respond and how strongly.

We asked three questions. (1) How much of the transcriptional variation in pooled spaceflight data is a flight response shared by all studies? (2) Do flight signatures replicate between studies, and does measurement platform matter? (3) Do CFD-predicted gas-exchange quantities explain differences between studies once study structure is respected?

Results

Data and harmonisation

We analysed 307 expression samples (153 flight, 154 ground control) from 11 OSDR studies: 213 RNA-seq samples from 7 studies and 94 microarray samples from 4 studies (Fig. 1). Studies were grouped into four hardware classes: BRIC, CARA, VEGGIE and EMCS. APEX-TAGES (OSD-7) flight samples were grouped with BRIC, but its ground controls were not sealed; EMCS (OSD-314) has no CFD scenario, so it enters the expression analyses but not the CFD analyses. BRIC ground controls used the sealed flight hardware, and CARA and VEGGIE ground controls were grown in the ISS Environmental Simulator chamber at Kennedy Space Center. CARA is represented by a single study (OSD-120).

We rebuilt the harmonised expression matrix from the 11 OSDR processed files and reproduced the original limma-voom fit: 25,453 genes, inter-study correlation 0.7327 against 0.7324, and t-statistic correlation 0.999. The original pipeline joined the platforms on gene identifiers and set genes not measured in a study to zero. We therefore repeated the main analyses without zero-filling, and with RNA-seq and microarray modelled separately (Methods).

The flight response is largely study-specific

We partitioned each gene's variance with every factor as a random effect and study nested in hardware (Fig. 2).

- **Study** explained 52.9% of the average gene's variance in the pooled data (17,129 genes measured in all studies) and 44.3% in the RNA-seq studies alone (24,774 genes).
- **A flight effect common to all studies** explained 0.3% and under 0.01%.
- **Study-specific flight effects** explained 7.1% and 11.0%, and hardware-specific flight effects 4.4% and 4.2%.

Only 7 of 17,129 genes (none of 24,774 in RNA-seq) had more than 10% of their variance in the common flight effect, against 3,889 and 8,848 in flight × study. Spaceflight changes expression substantially within experiments, but mostly in ways that differ between them.

Pooled flight contrasts depend on the reference hardware

In a factorial limma-voom model ($\sim$ flight × hardware + organ + light + ecotype, with study as a correlated block), the default flight coefficient is the flight effect in the reference hardware, BRIC: 11,797 genes at FDR < 0.05. The flight effect averaged over the four hardware classes involved 2,981 genes (254 with $|\log_2 \mathrm{FC}| > 1$), and the two estimates agreed only weakly across genes ($r = 0.21$). Restricting to genes measured in every study gave 10,915 and 2,361 genes. Which hardware is taken as the reference therefore largely determines what a pooled "flight effect" contains.

Flight signatures replicate between BRIC RNA-seq studies, not with microarray studies

Within BRIC, the RNA-seq and microarray flight effects were uncorrelated ($r = -0.05$). Platform is confounded with study and organ, so we estimated the flight effect separately in each study × organ unit and compared units from different studies (Fig. 3).

- The three BRIC RNA-seq studies agreed with one another (median $r = 0.40$).
- The BRIC microarray studies agreed neither with each other (median $r = 0.01$) nor with the RNA-seq studies (-0.03; whole seedlings only, -0.04).

- The microarray units detected a median of 5.5 flight genes at FDR < 0.05, against a median of 7,381 in the BRIC RNA-seq units.

The platform split is therefore weak or absent flight signal in the microarray studies, not an opposite response. These are the earlier flights (BRIC, and APEX-TAGES for OSD-7), so platform and flight era cannot be separated. Across hardware, even RNA-seq studies agreed poorly (median $r = 0.09$).

CFD covariates do not explain which studies respond

Each CFD covariate is set by hardware class, gravity, organ scale and ground-control enclosure, so neither covariate can be separated from study.

- g_{bl} relabels the flight × organ-scale cells (variance inflation factor 14.2).
- **Carbon supply** is a hardware label once BRIC ground controls are given the sealed-dish value: study membership explains 86% of its variance, and it varies within only two studies, OSD-7 (unsealed ground controls) and OSD-120 (CARA), where it coincides with the flight contrast.

At the pathway level, we scored 28 curated pathways within each study on each platform's own log scale. These scores avoid zero-filling and count each gene once. We then fitted a within-study model with cluster-robust standard errors by study.

- **Flight:** no pathway showed a flight effect at FDR < 0.05 (Fig. 4); three confidence intervals exclude zero before multiple-testing correction. A model restricted to the five studies with in-dish covariates flagged one pathway (ethylene signalling, 4 residual degrees of freedom), which the all-study model does not replicate.
- **Organ scale:** three pathways differed between rosette and leaf scale: ethylene biosynthesis, jasmonate signalling and gravitropism signalling.
- **CFD dose-response:** a meta-regression of study-level flight effects on the CFD-predicted boundary-layer penalty (13 study × scale units, Knapp–Hartung inference; EMCS has no CFD penalty) found no pathway at FDR < 0.05.
- **Power:** a planted flight effect of 0.25 or 0.5 within-study SD was detected in 8/28 and 15/28 pathways, so these tests rule out only large effects.

Inside the flight dish, gravity matters less than open-air models suggest

LunarLeaf-CFD resolves the leaf boundary layer at 0.288 mm in two dimensions, with buoyancy driven by CO_2, O_2 and H_2O concentration differences (Fig. 5).

- **Open box:** a leaf's g_{bl} falls from 0.997 at 1 g to 0.443 $\mathrm{mol\,m^{-2}\,s^{-1}}$ at 0 g, a 2.25-fold penalty (LunarLeaf-CFD T13).
- **Real dishes:** inside the BRIC 60 mm and CARA 120 mm dishes, the shallow headspace keeps convection weak even at 1 g, and the penalty shrinks to 1.4-fold in the taped CARA dish and about 1.8-fold in the sealed BRIC dish, whose 1 g value is indicative only (T19).
- **At 0 g:** both dishes give nearly the same conductance to their own air (0.185 and 0.186).
- **Sealed BRIC dish:** with photosynthesis feedback, it runs out of CO_2 in about 3.5 min at either gravity (T20). With humidity-limited transpiration, its 1 g and 0 g boundary layers converge within minutes (T22).

In the sealed hardware, ground controls are therefore carbon-limited too, and the gravity contrast in gas exchange is small. That is consistent with the absence of a boundary-layer dose-response in the transcriptomes.

Exploratory analyses

The original pipeline also deconvolved the bulk samples against the Shahan root atlas (GSE152766) [Shahan et al., 2022] with a scVI model [Lopez et al., 2018], and inferred miRNA activity from target-gene expression. Both were run on the zero-filled matrix; the deconvolution uses a root-only reference for mostly non-root samples. The miRNA analysis covered 19 miRNAs across 34 contrasts, and no target set was enriched among differentially expressed genes (all FDR = 1). We report these analyses in the supplementary tables but do not interpret them.

Discussion

The pooled OSDR data contain a large flight response within experiments but very little that is shared between them. A flight effect common to all studies explains under 1% of gene variance, against 7–11% for study-specific flight effects. Pooled contrasts are therefore fragile: the default "flight effect" of a factorial model is the effect in the reference hardware, and an estimate averaged over hardware differs substantially. Within one hardware, the RNA-seq studies do share a flight signature. This suggests that common biology exists, but is masked by differences in hardware, tissue, platform and flight.

The microarray studies contribute almost no flight signal. Whether this reflects the platform or the earlier flights cannot be decided with these data, but it means that pooling the two platforms dilutes rather than strengthens a flight signature.

The CFD covariates did not explain which studies respond. With so few independent studies per hardware, any hardware-level covariate is confounded with study by construction. The CFD itself also predicts a small effect: inside the flight dishes, the gravity penalty on leaf gas exchange is modest. In sealed BRIC dishes, carbon limitation is a property of the hardware, shared by flight and ground. Hardware atmospheres may well shape plant responses, but testing that needs replicated flights in common hardware, with ground controls in identical enclosures, rather than covariates applied after the fact to heterogeneous experiments.

Limitations.

- Study is nested in hardware, and CARA is a single study.
- APEX-TAGES flight samples share the BRIC CFD scenario, and EMCS has none.
- The positive controls show that the pathway tests can rule out only large effects.
- The CFD is two-dimensional, its humidity saturation level is an estimate, and it is not yet validated against in-flight measurements.

Methods

Data

Processed expression files for 11 OSDR studies were downloaded from the OSDR biodata API. For the 7 RNA-seq studies these were the GeneLab normalised counts and RSEM expected counts; for the 4 microarray studies, the normalised probe-set intensities. A manifest with source URLs and SHA-256 checksums is kept with the downloaded data (outside the repository). Sample metadata (ecotype, organ, flight, light, hardware) follows the original OSDR harmonisation; organ was mapped to leaf or rosette CFD scale.

Harmonisation

Microarray probe sets were mapped to AGI identifiers and averaged per gene. The original matrix was an outer join of all studies with unmeasured genes set to zero, and was rebuilt to reproduce the stored fit. Two corrected variants were analysed:

- **complete case:** genes measured in every study (17,129);
- **platform-separated:** RNA-seq modelled with voom on RSEM expected counts (TMM normalisation [Robinson & Oshlack, 2010], low-count filtering), and microarray with limma-trend on the published $\log_2$ intensities.

Differential expression

Models were fitted with limma-voom [Law et al., 2014, Smyth, 2005], with study as a correlated block estimated by duplicateCorrelation [Smyth et al., 2005].

- **Factorial model:** $\sim$ flight $\times$ hardware + organ + light + ecotype. The flight effect averaged over hardware was tested as a contrast giving each hardware equal weight.
- **CFD-covariate model:** replaced hardware with g_{bl} and carbon supply. Collinearity was assessed with variance inflation factors and with the variance of each covariate explained by study.

False discovery rates used the Benjamini–Hochberg procedure [Benjamini & Hochberg, 1995].

Variance partitioning

Variance components were estimated per gene with variancePartition [Hoffman & Schadt, 2016], using the formula $\sim$ (1|hardware) + (1|study) + (1|flight) + (1|flight:hardware) + (1|flight:study) + (1|organ) + (1|light) + (1|ecotype). It was applied to voom-weighted complete-case data and to the RNA-seq studies alone. The CFD covariates were omitted: they are set by hardware, gravity and scale and are confounded with study (canonical correlation with study 0.93 for carbon supply and 0.50 for g_{bl}; hardware itself 1.00).

Platform concordance

The flight effect was estimated separately in each study $\times$ organ unit with at least two flight and two ground samples. The model was flight plus ecotype and light where these varied within the unit (voom for RNA-seq, limma-trend for microarray). Concordance was the Pearson correlation of $\log_2$ fold changes over shared genes, summarised for pairs of units from different studies.

Pathway scores and models

The 28 curated gene sets were checked against UniProt. Repeated entries were removed, one misassigned gene was removed from the heat-shock set, and the heat-shock transcription factors and HSP101 were added. Scores were computed within each study as the mean within-study z-score of member genes. RNA-seq expression was taken as $\log_2$ counts per million, and microarray as published $\log_2$ intensities.

Each pathway score was modelled as $\sim$ study + flight $\times$ scale, with effect coding. Standard errors were cluster-robust (CR1) by study [Cameron & Miller, 2015], with t tests on $G - 1$ degrees of freedom; terms identified within fewer than two studies were not tested. Study-level flight effects were meta-regressed with Knapp–Hartung inference [Knapp & Hartung, 2003] on the predicted penalty: the increase in the magnitude of the mean leaf-surface CO_2 gap from ground to flight in each study $\times$ scale unit (flight values from the LunarLeaf-CFD hardware

scenes, T10, whose BRIC and CARA runs had not reached steady state; ground values from the open-box scene). Positive controls added a known flight effect to every pathway and recorded detection.

CFD

Boundary-layer quantities come from LunarLeaf-CFD, a two-dimensional D2Q9 lattice-Boltzmann solver with D2Q5 transport of CO_2, O_2 and H_2O and concentration-driven buoyancy. Leaf and rosette scenes were run to steady state (T13). The real BRIC and CARA dish cross-sections were modelled with the leaf 1 mm above the agar (T19). Sealed-dish runs added CO_2-limited photosynthesis (T20) and humidity-limited transpiration (T22). Covariates were assigned per sample by hardware class, gravity (flight = 0 g) and organ scale; APEX-TAGES ground controls, which were not sealed, take the unsealed carbon value, and EMCS samples have no CFD covariates.

Code and data availability

All code, tables and figures are at https://github.com/dr-richard-barker/Airflow_omics. The CFD model is at https://github.com/dr-richard-barker/LunarLeaf-CFD. Expression data are from NASA OSDR (https://osdr.nasa.gov). The full analysis record, with every number in this manuscript, is `tables/reanalysis_2026-10/REANALYSIS.md`. Archival DOI: TODO.

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Author contributions

R.B. conceived the study, developed the CFD model, performed all analyses, and wrote the manuscript.

Competing interests

The author declares no competing interests.

Supplementary Tables

All in `tables/reanalysis_2026-10/`:

- Table S1: differential expression for the refit contrasts (`gene_level/de_summary_v2.tsv` and `de_v2_*.tsv.gz`)
- Table S2: variance partition per gene and summary (`followups/variance_partition_*`)
- Table S3: platform concordance, per unit and per pair (`followups/platform_concordance_*`)
- Table S4: rescored pathway scores and within-study models (`followups/pathway_*`)
- Table S5: meta-regression units and results (`followups/rescored/`)

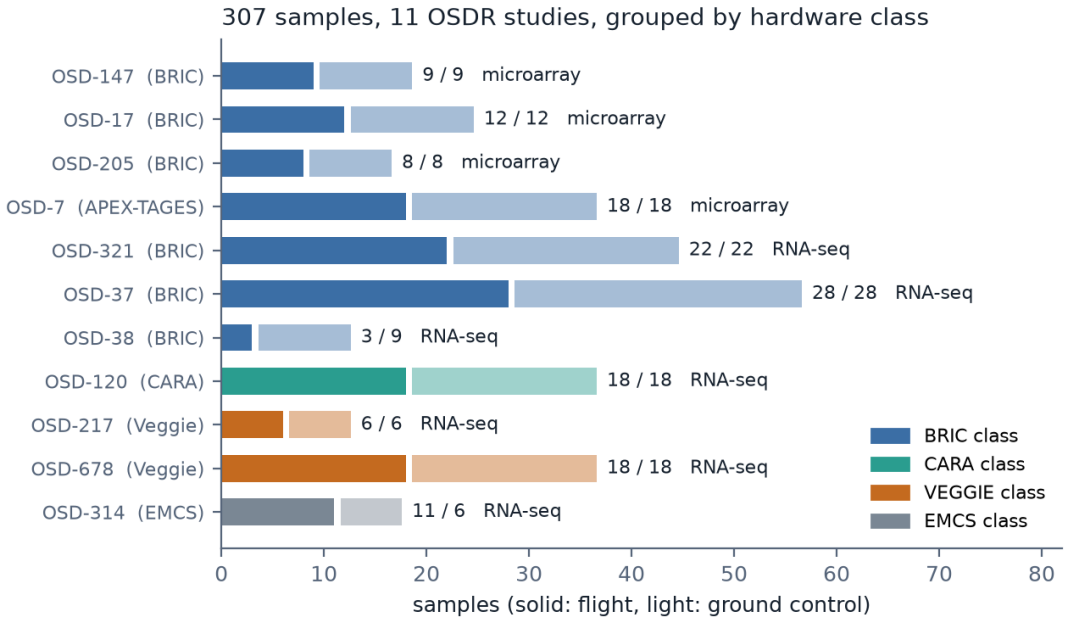


Figure 1: **Study design.** Flight (solid) and ground-control (light) samples in each of the 11 OSDR studies, coloured by hardware class and labelled by platform. APEX-TAGES (OSD-7) flight samples are grouped with BRIC; EMCS (OSD-314) has no CFD scenario. Source: `expression_metadata_with_cfd_v2.tsv`.

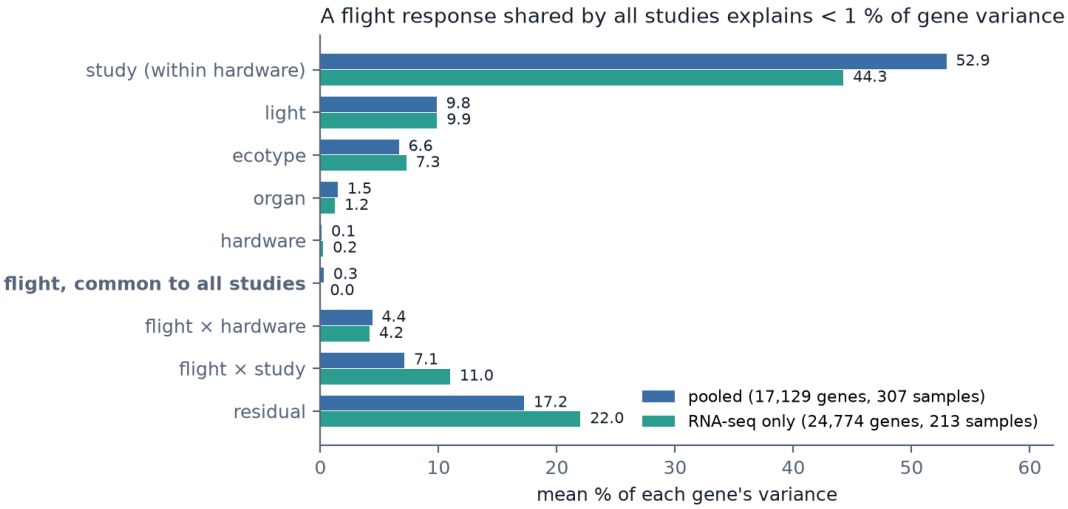


Figure 2: **The flight response is largely study-specific.** Mean percent of each gene's variance explained by each component, for the pooled data (genes measured in all studies) and the RNA-seq studies alone. Every factor is a random effect; study is nested in hardware. Source: REANALYSIS §12.

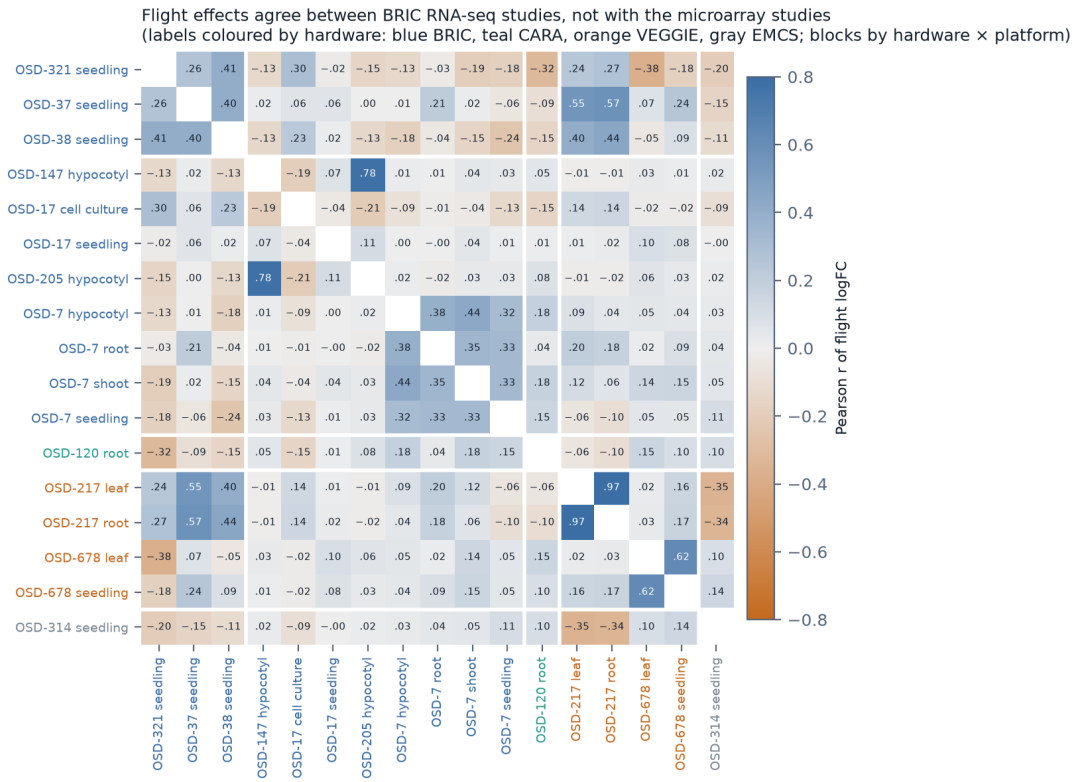


Figure 3: **Agreement of flight effects between studies.** Pearson correlation of flight $\log_2$ fold changes between study × organ units, grouped by hardware and platform. BRIC RNA-seq units agree (median $r = 0.40$ between studies); BRIC microarray units agree neither with each other nor with RNA-seq. Source: REANALYSIS §13.

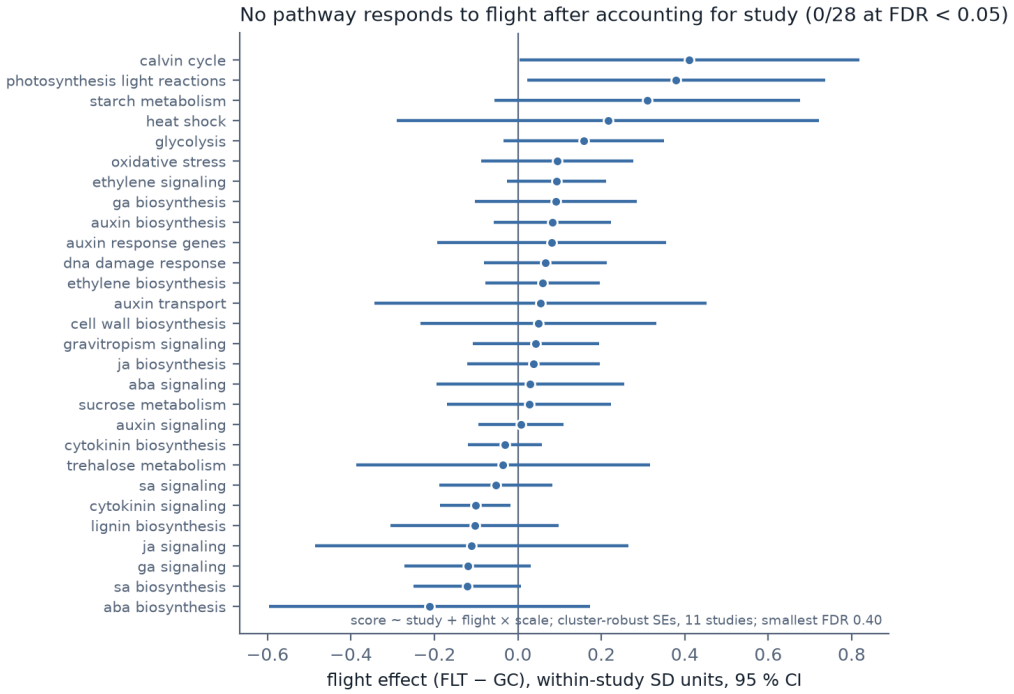


Figure 4: **No pathway-level flight effect.** Flight effect (flight minus ground) per pathway with 95% confidence intervals, from a within-study model with cluster-robust errors by study (11 studies), on scores computed within each study without zero-filling. Intervals are not adjusted for multiple testing; no pathway reaches FDR < 0.05. Source: REANALYSIS §10.

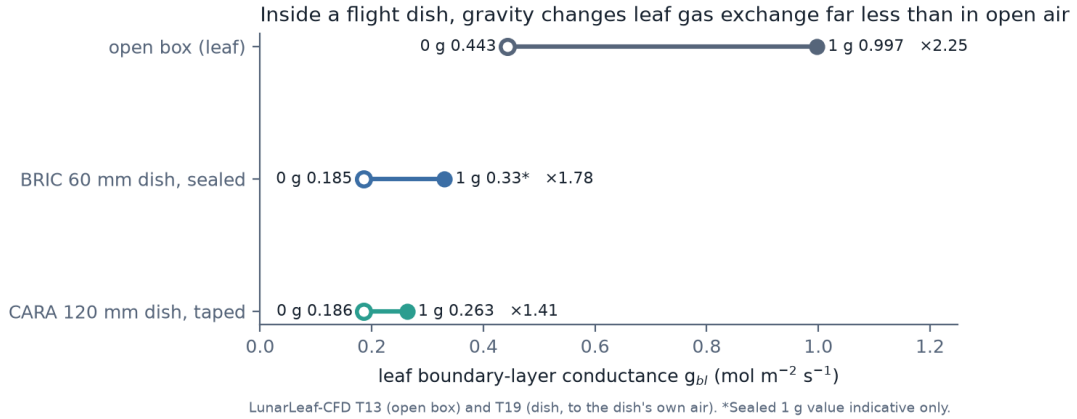


Figure 5: **Gravity penalty on leaf gas exchange, open air versus flight dish.** Leaf boundary-layer conductance at 1 g (filled) and 0 g (open) in an open box and inside the BRIC and CARA dishes, with the 1 g/0 g ratio. The sealed-BRIC 1 g value is indicative only. Source: LunarLeaf-CFD T13 and T19.

- Table S6: CFD covariates per sample (`expression_metadata_with_cfd_v2.tsv`)
- Table S7: exploratory deconvolution and miRNA inference from the original pipeline (`tables/deconvolution_proportions.tsv`, `tables/mirna_activity_inference.tsv`)

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